07

Lecture 07

LTE Scheduling

زمان‌بندی در LTE

2G — 3G — **4G** — 5G — 6G

COURSE PART
Part II – 4G access and core (Lectures 5–13)

DIFFICULTY
Intermediate to advanced

ESTIMATED STUDY TIME
8–10 hours (3.5 h reading, 4–6 h exercises incl. simulation)

PREREQUISITES
Lectures 5–6; Shannon capacity; Rayleigh fading; basic optimisation

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Original lecture notes for the course *Advanced Communication Networks*. The lecture sequence follows the course syllabus (26 lectures, 2G to 6G). Equations use identical notation in the English and Persian editions. Companion edition: Persian (`lecture-07-fa.pdf`).

1 Learning objectives

After studying this lecture you should be able to:

- **LO1** **Describe** the LTE resource grid – frame, subframe, slot, resource element, resource block – and **count** the resources of any LTE bandwidth.
- **LO2** **Explain** downlink and uplink scheduling signalling: DCI, grants, BSR, PHR, CQI.
- **LO3** **Apply** link adaptation: map CQI to modulation and coding, and **analyse** outer-loop link adaptation (OLLA).
- **LO4** **Analyse** HARQ with eight stop-and-wait processes and compute its throughput effect.
- **LO5** **Compare** round robin, maximum C/I and proportional fair scheduling and **derive** why PF maximises the sum of log-throughputs.
- **LO6** **Evaluate** the throughput–fairness–QoS trade-off with Jain's index and a simulation.

2 Prerequisite knowledge

You need the LTE protocol stack (Lecture 6), in particular the MAC layer's scheduling and HARQ roles, the Shannon capacity $C = B\log_2(1+\mathrm{SNR})$ and the fact that, under Rayleigh fading, the instantaneous SNR is exponentially distributed around its mean. Basic calculus (derivatives, Lagrange-style arguments) is used in Section 6.4.

Key terms. The vocabulary of this lecture with its Persian equivalents:

Term	Persian	Meaning in this lecture
Resource element (RE)	عنصر منبع	One subcarrier during one OFDM symbol.
Resource block (RB)	بلوک منبع	12 subcarriers (180 kHz) × one slot.
TTI	بازهٔ زمانی ارسال	Scheduling interval: one 1-ms subframe in LTE.
DCI	اطلاعات کنترلی فروسو	Scheduling message on the PDCCH: resources, MCS, HARQ.
CQI	نشانگر کیفیت کانال	UE report: highest MCS decodable at 10% BLER.
MCS	طرح مدولاسیون و کدگذاری	Modulation order + code rate chosen by the scheduler.
BSR / PHR	گزارش وضعیت بافر / حاشیهٔ توان	Uplink reports: queued data / remaining power.
HARQ	درخواست تکرار خودکار ترکیبی	Fast retransmission with soft combining.
Proportional fair (PF)	منصفانهٔ تناسبی	Scheduler maximising $\sum \log R_k$.
Multi-user diversity	تنوع چندکاربره	Gain from serving users at their fading peaks.

3 Introduction and motivation

Every millisecond an LTE eNB must answer one question: *who gets which resource blocks, with which modulation and coding?* The answer determines everything users perceive – throughput, delay, fairness – and everything the operator pays for, because spectrum is the most expensive resource of a mobile network.

The problem is hard because the channel of each user fluctuates (fast fading, mobility, interference), traffic is bursty, and the objectives conflict: maximising total throughput favours users close to the base station, while fairness and QoS guarantees require serving users at the cell edge. LTE gives the eNB a fine-grained tool – 1 ms in time, 180 kHz in frequency – and leaves the scheduling *algorithm* to the vendor. 3GPP standardises the signalling (what the UE reports and how grants are sent), not the decision. This lecture covers both: the standard mechanisms and the classic algorithms with their mathematical foundations.

4 Fundamental concepts

4.1 The resource grid

In the time domain an LTE radio frame lasts 10 ms and contains ten 1 ms *subframes*, each made of two 0.5 ms *slots* of 7 OFDM symbols (normal cyclic prefix) [2]. In the frequency domain, subcar-

riers are spaced $\Delta f = 15\,\text{kHz}$. A *resource element* is one subcarrier for one symbol; a *resource block* is 12 subcarriers (180 kHz) for one slot (84 REs). The scheduler normally assigns *RB pairs* (both slots of a subframe, 168 REs) every TTI of 1 ms.

Table 1. LTE channel bandwidths and resource blocks [2].

Channel bandwidth [MHz]	1.4	3	5	10	15	20
Resource blocks N_{RB}	6	15	25	50	75	100
Occupied bandwidth [MHz]	1.08	2.7	4.5	9.0	13.5	18.0

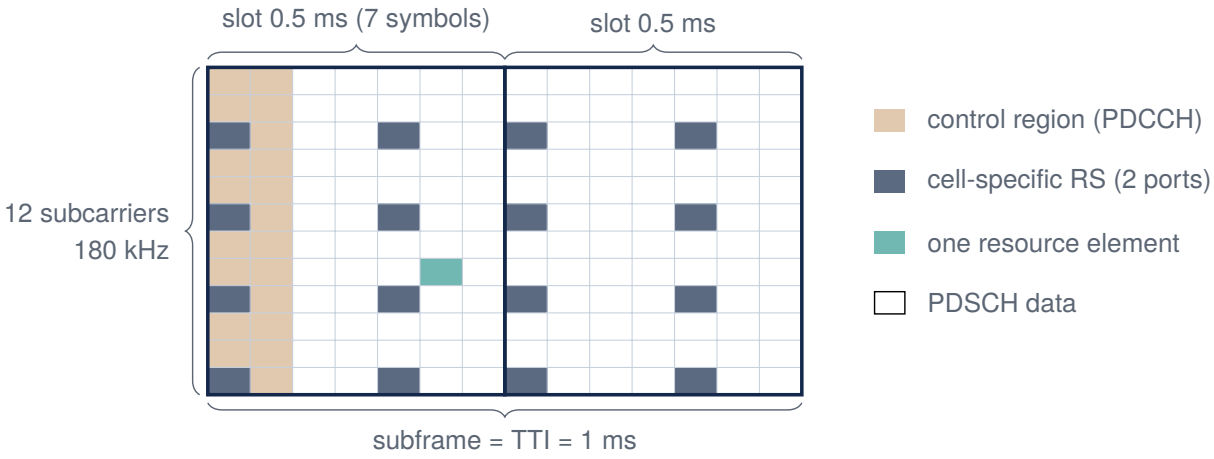


Figure 1. One downlink RB pair (12 subcarriers × 14 symbols) with a two-symbol control region and two-port cell-specific reference signals.

4.2 What the scheduler knows

- **Channel quality**: CQI reports (wideband or per subband), rank and precoder indicators (RI, PMI) in the downlink; sounding reference signals (SRS) in the uplink.
- **Traffic**: downlink buffers are in the eNB; uplink buffers are reported by the UE in *buffer status reports* (BSR), per group of logical channels.
- **QoS**: the QCI, guaranteed bit rate (GBR) and priority of each bearer (Lecture 22).
- **UE power**: *power headroom reports* (PHR) tell how much more power the UE could use, so the uplink scheduler does not grant more RBs than the UE can power.

5 Detailed technical explanation

5.1 Downlink scheduling

For each scheduled UE the eNB sends a *downlink control information* (DCI) message on the PDCCH in the control region of the same subframe [3]. A DCI contains the resource allocation, the MCS index, the HARQ process number, the new-data indicator, the redundancy version and a power-control command for PUCCH. Three resource allocation types trade signalling size against flexibility: type 0 (bitmap of RB groups), type 1 (RBs within a subset) and type 2 (a contiguous

set, compact). The UE decodes the PDSCH in the same subframe and returns ACK/NACK four subframes later.

5.2 Uplink scheduling

The uplink differs in three ways. (i) The eNB does not see the UE's buffer: the UE sends a *scheduling request* (SR) on PUCCH, receives a small grant, and then reports its buffer with a BSR. (ii) Grants are sent on the PDCCH four subframes before the PUSCH transmission (FDD). (iii) SC-FDMA requires the RBs of one UE to be *contiguous* in frequency to keep the low peak-to-average power ratio, which limits frequency-selective scheduling. Uplink HARQ is *synchronous*: retransmissions occur exactly 8 ms later, acknowledged on the PHICH.

5.3 CQI and link adaptation

The UE reports a 4-bit CQI: the highest index whose modulation and code rate would be received with a transport-block error probability not exceeding 10% [3]. Table 2 lists the standard table. The eNB maps the CQI (after correction, see OLLA below) to an MCS index and a transport block size.

Table 2. LTE 4-bit CQI table (TS 36.213, Table 7.2.3-1). Efficiency is in information bits per resource element.

CQI	Modulation	Code rate×1024	Efficiency	CQI	Modulation	Code rate×1024	Efficiency
1	QPSK	78	0.1523	9	16QAM	616	2.4063
2	QPSK	120	0.2344	10	64QAM	466	2.7305
3	QPSK	193	0.3770	11	64QAM	567	3.3223
4	QPSK	308	0.6016	12	64QAM	666	3.9023
5	QPSK	449	0.8770	13	64QAM	772	4.5234
6	QPSK	602	1.1758	14	64QAM	873	5.1152
7	16QAM	378	1.4766	15	64QAM	948	5.5547
8	16QAM	490	1.9141	0	out of range		

Note that the efficiency is simply $Q_m \times (\text{code rate})$: for CQI 15, $6 \times 948/1024 = 5.5547$ bits per RE. Wideband CQI describes the whole carrier; subband CQI (per group of RBs) enables frequency-selective scheduling but costs uplink overhead.

5.4 HARQ

LTE uses eight parallel stop-and-wait HARQ processes in FDD (Figure 2): while process 2 waits for its acknowledgement, processes 3–8 use the following subframes, so the channel never idles. With *Chase combining* the retransmission repeats the same coded bits and the receiver adds the signal energies; with *incremental redundancy* it sends new parity bits, lowering the effective code rate. Typical operation targets a 10% first-transmission BLER and a residual BLER after HARQ of about 10^{-3} or below; RLC ARQ (Lecture 6) catches the rest.

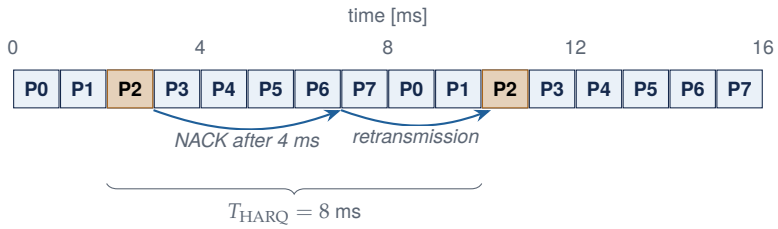


Figure 2. Eight HARQ processes in FDD. A NACK for process P2 arrives four subframes after the transmission; the retransmission occurs eight subframes after the original.

5.5 Scheduling algorithms

Round robin (RR). Users take turns regardless of channel quality. Perfectly fair in *time*, but it ignores the channel and wastes good fading peaks.

Maximum C/I (Max-rate). In each TTI and RB, serve the user with the highest instantaneous rate $r_k(t)$. Maximises cell throughput but starves cell-edge users.

Proportional fair (PF). Serve the user with the largest ratio of instantaneous rate to its own average throughput [5]:

$$k^*(t) = \arg\max_k \frac{r_k(t)}{\bar{R}_k(t)}, \qquad \bar{R}_k(t+1) = \left(1 - \tfrac{1}{t_c}\right)\bar{R}_k(t) + \tfrac{1}{t_c}\, r_k(t)\, \mathbb{1}\{k = k^*(t)\}, \tag{1}$$

where t_c is the averaging window in TTIs. Each user is served when its channel is *good relative to its own average*, so cell-edge users still get their peaks.

QoS-aware scheduling. Real schedulers first satisfy GBR bearers (e.g. VoLTE, often with semi-persistent scheduling), weight metrics by QCI priority and packet delay budget, and use PF-like rules for the remaining best-effort traffic.

6 Mathematical models

6.1 From CQI to data rate

For a UE allocated N_{RB} RB pairs with $N_{\mathrm{RE}}^{\mathrm{data}}$ data REs each and an efficiency η_{CQI} (bits/RE) from Table 2, the rate per layer is

$$R = \eta_{\mathrm{CQI}} \, \frac{N_{\mathrm{RB}}\, N_{\mathrm{RE}}^{\mathrm{data}}}{T_{\mathrm{sf}}}. \tag{2}$$

Example: CQI 9 ($\eta = 2.4063$), 50 RBs, 132 data REs (two control symbols, two CRS ports): $R = 2.4063 \times 50 \times 132/10^{-3} = 15.9\,\mathrm{Mbit/s}$.

6.2 Outer-loop link adaptation

Problem. CQI reports are delayed and biased (estimation errors, interference changes); if the eNB used them directly the actual BLER would drift from the 10% target.

Model. The eNB adds an offset Δ (dB) to the reported SINR before choosing the MCS. After each ACK it raises Δ by δ_{up}; after each NACK it lowers it by δ_{down}.

Derivation. In steady state the expected change is zero. If p is the BLER at the operating point,

$$(1-p)\,\delta_{\text{up}} - p\,\delta_{\text{down}} = 0 \quad\Longrightarrow\quad p = \frac{\delta_{\text{up}}}{\delta_{\text{up}} + \delta_{\text{down}}}. \tag{3}$$

Choosing $\delta_{\text{up}} = \delta_{\text{down}}\, p_{\text{t}}/(1-p_{\text{t}})$ therefore drives the BLER to the target p_{t} regardless of the CQI bias. For $p_{\text{t}} = 0.1$ and $\delta_{\text{down}} = 0.5\,\text{dB}$, $\delta_{\text{up}} = 0.056\,\text{dB}$.

Interpretation. OLLA is a stochastic control loop: small steps give an accurate BLER but slow convergence (about $1/\delta$ steps), large steps react fast but jitter the MCS.

6.3 HARQ throughput

If the first transmission fails with probability p_1 and a retransmission (after combining) fails with probability $p_2 < p_1$, the mean number of transmissions per transport block (neglecting a third attempt's failures) is $\bar{N} \approx 1 + p_1 + p_1 p_2$, and the throughput efficiency is $1/\bar{N}$. With $p_1 = 0.1$ and $p_2 = 0.01$: $\bar{N} = 1.101$, efficiency 90.8%. A 10% BLER target costs only about 9% of throughput but buys a much higher MCS than a 1% target would allow – the reason LTE operates at 10%.

6.4 Why proportional fair is "proportionally fair"

Claim. Among all scheduling policies, PF (asymptotically, for long averaging windows) maximises the utility

$$U = \sum_{k=1}^{K} \log \bar{R}_k. \tag{4}$$

Argument. Consider giving the current TTI to user k. To first order it increases $\bar{R}_k$ by $\epsilon\, r_k(t)$ (with $\epsilon = 1/t_{\text{c}}$), changing the utility by

$$\Delta U \approx \frac{\partial U}{\partial \bar{R}_k}\,\epsilon\, r_k(t) = \epsilon\,\frac{r_k(t)}{\bar{R}_k}. \tag{5}$$

The greedy choice that maximises the utility increase in every TTI is therefore exactly $\arg\max_k r_k(t)/\bar{R}_k$, i.e. Eq. (1). Because U is concave, greedy steps converge to its maximum. A useful property of the optimum is *proportional fairness*: any feasible change of the throughput vector satisfies $\sum_k \Delta\bar{R}_k/\bar{R}_k \le 0$ – no user can gain a percentage of throughput without others losing at least the same total percentage. Max C/I maximises $\sum \bar{R}_k$ instead, and RR equalises time shares.

6.5 Jain's fairness index

To compare schedulers we use [4]

$$J = \frac{\left(\sum_{k=1}^{K} \bar{R}_k\right)^2}{K \sum_{k=1}^{K} \bar{R}_k^2}, \qquad \frac{1}{K} \le J \le 1, \tag{6}$$

which equals 1 when all users get the same throughput and $1/K$ when one user gets everything.

6.6 Simulation: multi-user diversity

Set-up. `scripts/scheduler_sim.py` simulates one 10 MHz carrier (9 MHz occupied) given entirely to one user per TTI; four users with mean SNRs of 20, 12, 5 and 0 dB; independent Rayleigh

block fading per TTI; efficiency $\min(0.75\log_2(1+\mathrm{SNR}), 5.55)$ bit/s/Hz (an attenuated Shannon mapping capped at CQI 15); 20 000 TTIs; $t_c = 100$; fixed random seed.

Results. (Figure 3, Table 3)

Table 3. Simulated per-user and cell throughput [Mbit/s] and Jain's index.

Scheduler	UE1 (20 dB)	UE2 (12 dB)	UE3 (5 dB)	UE4 (0 dB)	Cell	J
Round robin	9.74	5.83	2.88	1.43	19.88	0.710
Prop. fair	12.97	8.34	4.60	2.50	28.40	0.761
Max C/I	35.73	4.18	0.21	0.01	40.13	0.311

Interpretation. PF gives *every* user more throughput than RR: each user still gets about a quarter of the TTIs, but they are its best TTIs. This is *multi-user diversity*, a gain that grows with the number of users and the fading depth. Max C/I raises the cell total by another 41% but essentially switches off UE3 and UE4. The model ignores frequency selectivity (subband scheduling adds further diversity), CQI delay and errors, and traffic dynamics (all users always have data – the "full buffer" assumption).

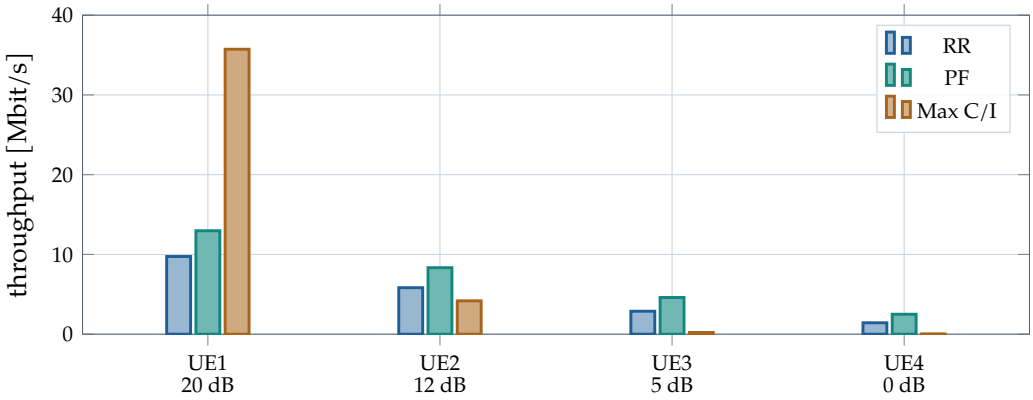


Figure 3. Per-user throughput under round robin, proportional fair and maximum C/I scheduling (simulation of Section 6.6).

7 Network architecture and diagrams

Scheduling is entirely a function of the eNB MAC; the only network-level inputs are the QoS parameters of each bearer, set by the core (QCI, ARP, GBR, MBR, AMBR) through S1AP when bearers are established (Lectures 9 and 22). Between eNBs, X2 *load information* messages allow inter-cell interference coordination: neighbours can agree to avoid scheduling cell-edge users on the same RBs. Figure 4 places the algorithms in the space of objectives.

8 Step-by-step procedures

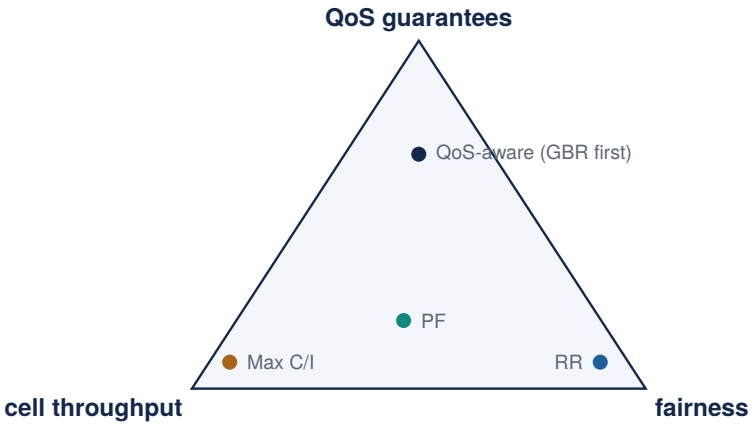


Figure 4. The scheduling trilemma. No algorithm maximises throughput, fairness and QoS at once; practical schedulers combine a QoS stage with a PF-like stage.

Procedure 1 • One TTI of downlink scheduling

1. Collect inputs: latest CQI/RI/PMI per UE, HARQ feedback received for subframe $n-4$, buffer sizes per bearer, QoS parameters.
2. Reserve resources for pending HARQ retransmissions (they have priority to keep delay low).
3. Serve GBR bearers that are below their guaranteed rate (e.g. VoLTE), using the MCS from corrected CQI (OLLA offset).
4. Rank the remaining backlogged UEs with the PF metric per RB group and allocate RBs.
5. For each scheduled UE choose MCS and transport block size; build the MAC PDUs (multiplexing logical channels by priority).
6. Encode the DCIs into PDCCH candidates (if the control region is full, defer UEs to the next TTI) and transmit PDCCH and PDSCH.
7. Update PF averages and OLLA offsets when ACK/NACK arrive four subframes later.

Procedure 2 • Uplink access for a UE with new data

1. New data arrives in an empty buffer; the UE triggers a BSR. Having no grant, it sends a scheduling request on its next PUCCH SR opportunity (or random access if none is configured).
2. The eNB sends a small grant (DCI format 0); four subframes later the UE transmits the BSR (and PHR) on PUSCH.
3. The eNB schedules the UE according to the reported buffer, its uplink SINR (from SRS/DMRS) and its power headroom, with contiguous RBs.
4. Each PUSCH transmission is acknowledged on PHICH after four subframes; non-adaptive retransmissions follow eight subframes after the original.

9 Worked examples

Worked example 1 • Counting resources

Problem. How many REs does a 15 MHz LTE carrier offer per second, and how many of them are PDSCH REs with a three-symbol control region and four CRS ports?
Solution. $75 \times 12 \times 14 \times 1000 = 12.6 \times 10^6$ RE/s. Data REs per RB pair: $168 - 36 - 16 = 116$ (four CRS ports use 24 REs, 8 of which lie in the three control symbols). Total $75 \times 116 \times 1000 = 8.7 \times 10^6$ RE/s, i.e. 69%.

Worked example 2 • OLLA step

Problem. An operator wants a 20% BLER target for a delay-tolerant service and uses $\delta_{\text{down}} = 1$ dB. Find δ_{up} and the offset drift if the actual BLER were 10%.
Solution. $\delta_{\text{up}} = 1 \times 0.2/0.8 = 0.25$ dB. At 10% BLER the expected change per transmission is $0.9 \times 0.25 - 0.1 \times 1 = +0.125$ dB: the offset rises (more aggressive MCS) until the BLER reaches 20%.

Worked example 3 • PF decision

Problem. Two UEs: UE A has average 10 Mbit/s and instantaneous 18 Mbit/s; UE B has average 2 Mbit/s and instantaneous 5 Mbit/s. Whom do Max C/I and PF serve?
Solution. Max C/I: UE A ($18 > 5$). PF: $18/10 = 1.8$ versus $5/2 = 2.5 \Rightarrow$ UE B, which is at a fading peak relative to its own average.

10 Technology comparison

Table 4. Scheduling and link adaptation across generations.

	HSDPA (3.5G)	LTE (4G)	NR (5G)
Scheduling unit	2 ms TTI, codes	1 ms TTI, RB pairs (180 kHz)	slot (1–0.125 ms) or mini-slot, RBs of $12\Delta f$
Frequency domain	none (CDMA)	per RB group (OFDMA)	per RB, bandwidth parts
Channel feedback	CQI	CQI, PMI, RI (subband)	CSI-RS based, beam reports
HARQ	NodeB, 6 processes	8 processes, 8 ms RTT (FDD)	up to 16 processes, flexible timing
Uplink	E-DCH (Rel-6)	SC-FDMA, contiguous RBs	CP-OFDM or DFT-s-OFDM, configured grants

Design trade-offs. Finer granularity enables better adaptation but costs control overhead (more DCIs, more CQI reports); frequent feedback improves accuracy but uses uplink capacity; longer PF windows favour throughput, shorter ones favour short-term fairness and delay. 5G extends the same principles with flexible numerology (Lecture 14).

11 Key takeaways

Key takeaways

- Frame 10 ms → subframe 1 ms → slot 0.5 ms (7 symbols); RB = 12 subcarriers × slot; $N_{\mathrm{RB}} = 6 \ldots 100$ for 1.4–20 MHz.
- 3GPP standardises signalling (DCI, CQI, BSR, PHR, HARQ timing), not the scheduling algorithm.
- CQI → efficiency $Q_m \times$ code rate; OLLA drives the BLER to $\delta_{\mathrm{up}} / (\delta_{\mathrm{up}} + \delta_{\mathrm{down}})$.
- Eight HARQ processes keep the channel busy; a 10% first-transmission BLER costs about 9%.
- Max C/I maximises $\sum \bar{R}_k$, PF maximises $\sum \log \bar{R}_k$, RR equalises time.
- Multi-user diversity: PF can beat RR for *every* user in fading channels.
- Practical schedulers: HARQ retransmissions first, then GBR/QoS, then PF for best effort.

12 Exercises

Exercise 1 • Level 1 – conceptual • Standard versus vendor

Which of the following are standardised by 3GPP and which are vendor implementation? CQI table, proportional fair metric, DCI format, OLLA step sizes, HARQ timing in FDD, priority of GBR bearers.

Exercise 2 • Level 1 – conceptual • Uplink contiguity

Explain why SC-FDMA requires contiguous RBs per UE and what this means for frequency-selective uplink scheduling.

Exercise 3 • Level 2 – numerical • Rate from CQI

A UE reports CQI 12 with rank 2 on a 20 MHz carrier (one control symbol, two CRS ports). It receives 40 RBs. Compute its rate with Eq. (2).

Exercise 4 • Level 2 – numerical • Jain's index

Verify the Jain indices of Table 3 for RR and Max C/I from the per-user values.

Exercise 5 • Level 3 – analytical • Two-user PF analysis

Two users have i.i.d. channels with the same distribution but user 1's rate is always twice user 2's ($r_1 = 2r_2$ in every TTI). Show that PF gives each user half of the TTIs, and compute the resulting throughputs relative to RR. What changes if the channels are independent?

Exercise 6 • Level 3 – analytical • Simulation study

Modify `scheduler_sim.py` to use 8 users with mean SNRs uniformly spaced between 0 and 21 dB and plot cell throughput and Jain's index versus $t_c \in \{10, 100, 1000\}$ for PF. Explain the trend.

Solutions

Solution to Exercise 1

Standardised: CQI table, DCI formats, HARQ timing in FDD (and the meaning of BSR/PHR). Vendor implementation: the PF metric (or any algorithm), OLLA step sizes, the prioritisation policy among bearers (constrained by the QoS parameters but not dictated in detail).

Solution to Exercise 2

SC-FDMA is DFT-spread OFDM: mapping the DFT output to contiguous subcarriers produces a single-carrier-like signal with low peak-to-average power ratio, allowing efficient power amplifiers in the handset. Non-contiguous mapping raises the PAPR. Hence the scheduler can choose *where* a UE's block lies (frequency-selective gain) but not pick scattered good RBs; Release 10 relaxed this with clustered allocation of two clusters.

Solution to Exercise 3

Data REs per RB pair: $168 - 12 - (16 - 4) = 144$. $R = 2 \times 3.9023 \times 40 \times 144/10^{-3} = 45.0\,\text{Mbit/s}$.

Solution to Exercise 4

RR: sum $= 19.88$, sum of squares $= 94.87 + 33.99 + 8.29 + 2.04 = 139.2$; $J = 395.2/(4 \times 139.2) = 0.710$. Max C/I: sum $= 40.13$, squares $\approx 1276.6 + 17.5 + 0.04 = 1294.2$; $J = 1610.4/(4 \times 1294.2) = 0.311$.

Solution to Exercise 5

With $r_1 = 2r_2$ always, $r_1/\bar{R}_1$ versus $r_2/\bar{R}_2$ compares $2r_2/\bar{R}_1$ with $r_2/\bar{R}_2$; equilibrium requires both to be chosen, which happens when $\bar{R}_1 = 2\bar{R}_2$, i.e. when each user gets half the TTIs – exactly RR, because the channels are perfectly correlated and there is no multi-user diversity to exploit. With independent channels PF picks each user at its relative peaks, and both throughputs exceed RR, as in Table 3.

Solution to Exercise 6

With mean SNRs $0, 3, \ldots, 21$ dB and 40 000 TTIs the script gives: $t_c = 10$: cell 29.8 Mbit/s, $J = 0.831$; $t_c = 100$: 32.8 Mbit/s, $J = 0.838$; $t_c = 1000$: 33.2 Mbit/s, $J = 0.839$ (RR: 21.8 Mbit/s, $J = 0.775$). A short window reacts to recent service and sometimes serves a user who is not at a relative peak; longer windows let PF wait for genuine peaks, so throughput rises and saturates once the window is much longer than the fading (here independent from TTI to TTI). The price of long windows in practice is short-term unfairness and larger delay jitter when channels are time-correlated.

13 Further reading

The physical channels and grid are specified in [2], CQI, DCI and HARQ procedures in [3] and MAC procedures (BSR, PHR, SR) in [1]. The proportional fair scheduler originated in CDMA-HDR [5]; its multi-user diversity interpretation is developed in Tse and Viswanath [7]. Jain's index is from [4]. Sesia et al. [6] cover LTE scheduling in depth.

References

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